Industrial Engineering and Operations Research

First Lecture by Prof. Uday S. Dixit, Department of Mechanical Engineering, IIT Guwahati



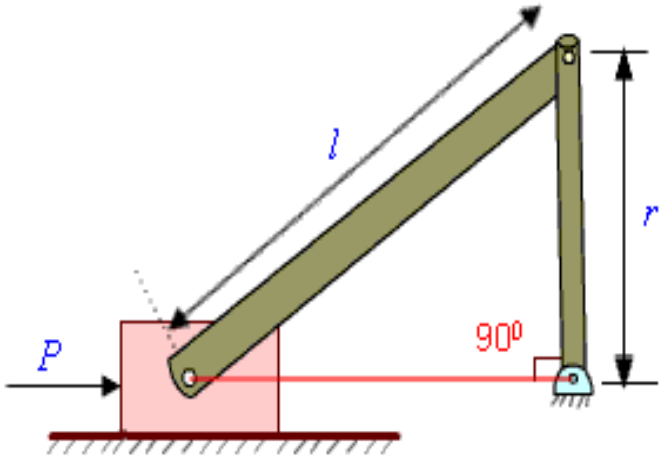
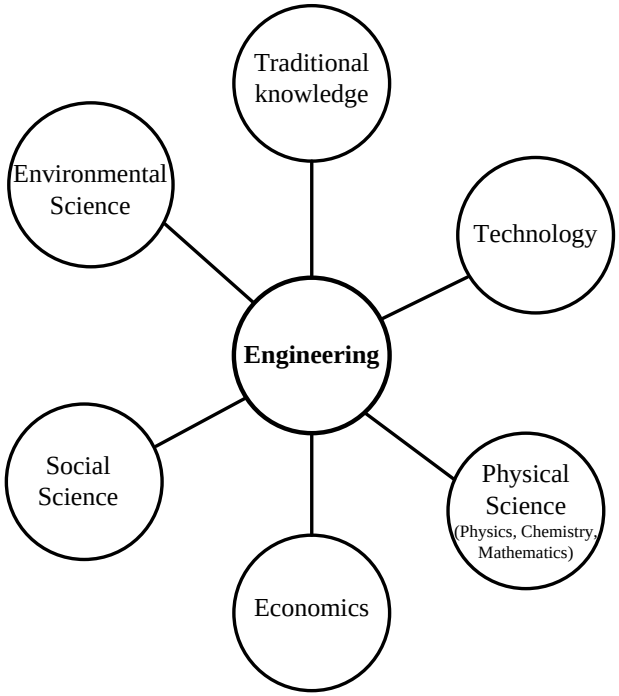
What is Engineering?

ENGINEERING is the profession in which a knowledge of the mathematical and natural sciences gained by study, experience, and practice is applied with judgment to develop ways to utilize economically the materials and forces of nature for the benefit of mankind.

(ABET definition)



https://commons.wikimedia.org/wiki/File:Expedition_36_flight_engineer_Chris_Cassidy.jpg

Science	Technology	Engineering
$F=ma$ $PV=mRT$ $E=mc^2$		

An ancient door safety technology



Engineering requires to use technology judiciously.



Mechanical Engineering

- **It deals with the design, construction, operation and maintenance of the machine.**
- **Machine is a device, comprising a number of interrelated parts, that makes mechanical work easier by overcoming a load at one end by the application of force at the other end.**

Mechanical Engineering: the oldest engineering

- **Pebble tools were first used by humans by more than 2 millions ago.**



https://en.wikipedia.org/wiki/Oldowan#/media/File:Canto_tallado_2-Guelmim-Es_Semara.jpg

Courtesy: Locutus Borg (Author)

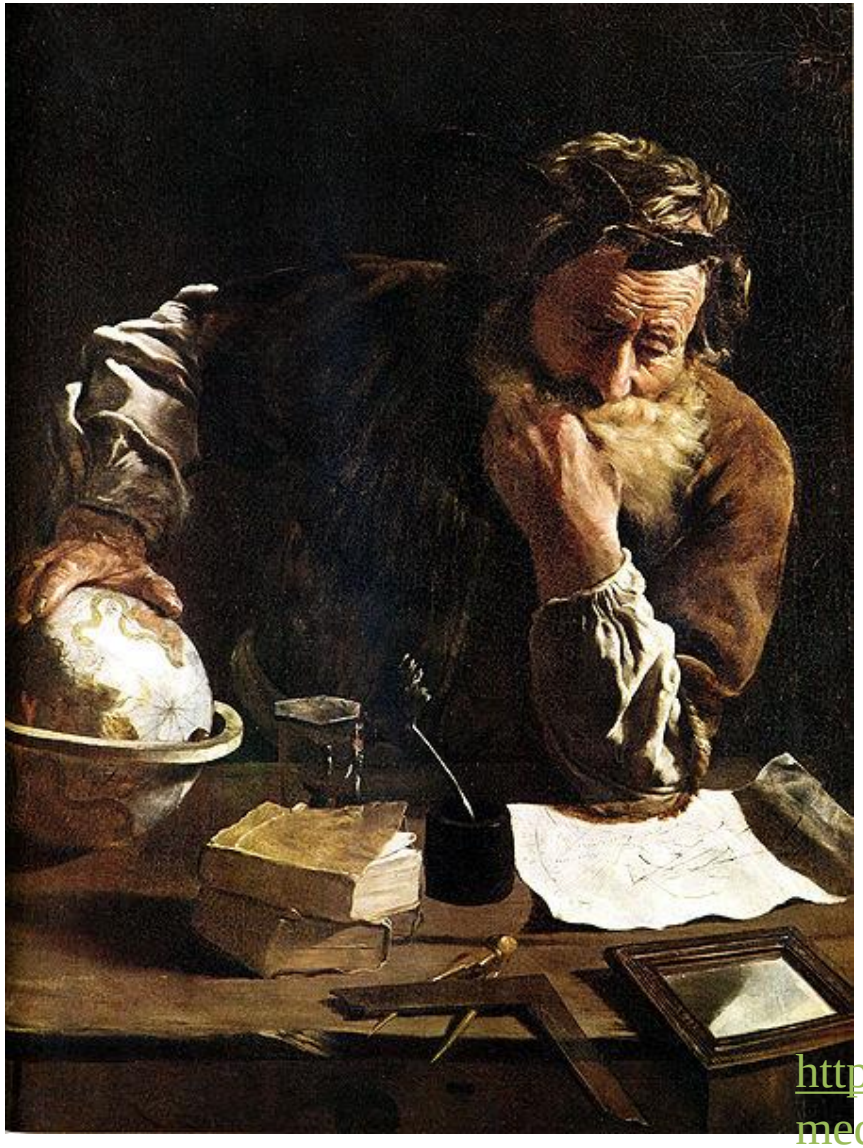
Potter's wheel: 3500 BC

Chariot found by excavation in Sanauli village near Baghpat in Uttar Pradesh (Circa 2000 BC)



https://www.indiatoday.in/india/story/indians-used-chariots-4-000-years-ago-asi-unearths-evidence-in-up-1251650-2018-06-06

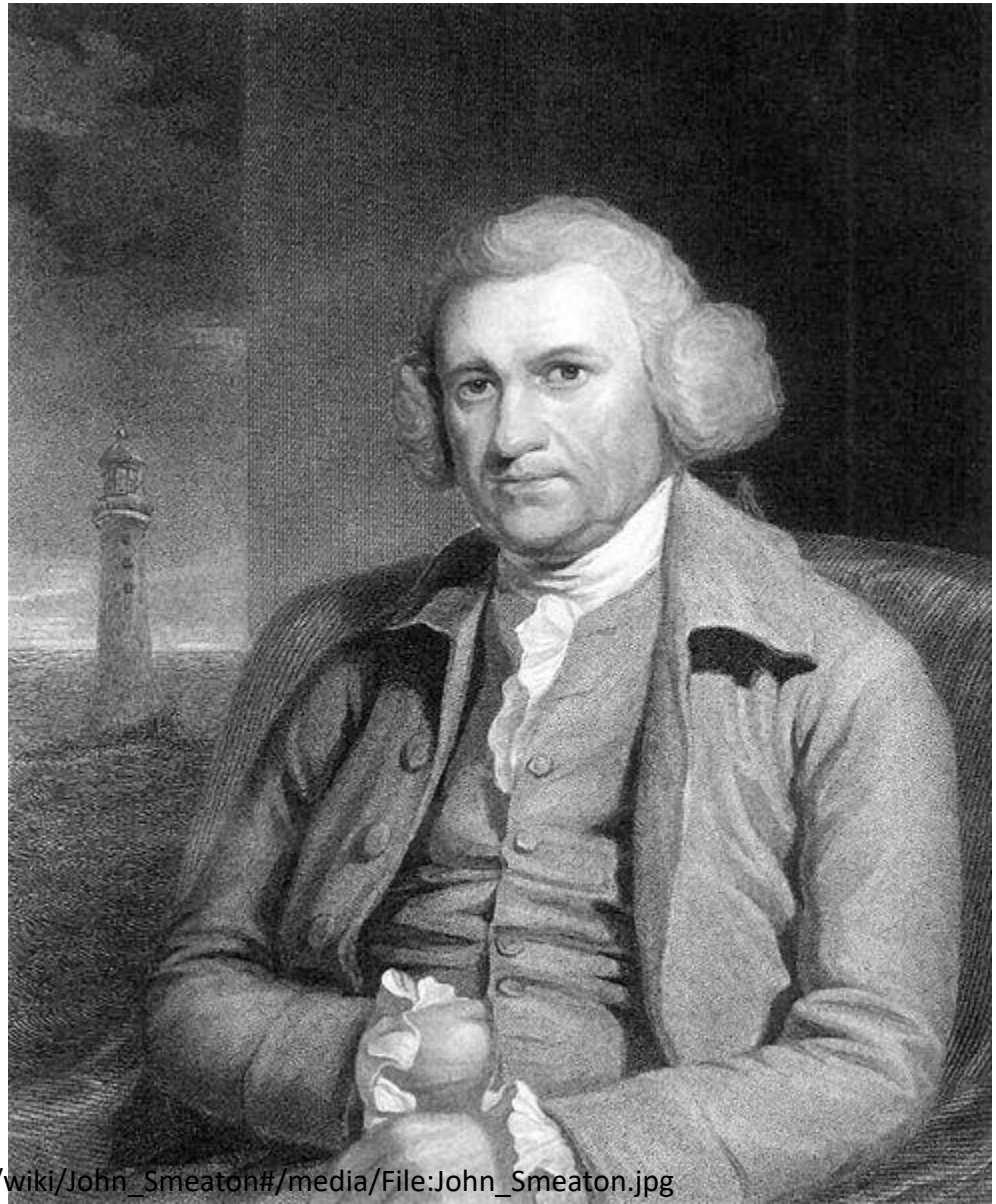
Archimedes (287-212 BC): the first (?) military engineer



https://en.wikipedia.org/wiki/Archimedes#/media/File:Domenico-Fetti_Archimedes_1620.jpg

Courtesy: Domenico Fetti (Painter)

John Smeaton (1724-1792), Father of Civil Engineering



https://en.wikipedia.org/wiki/John_Smeaton#/media/File:John_Smeaton.jpg

John Smeaton’s contributions to mechanical engineering

- Devised a water engine in 1761 and a watermill in 1767.
- Invented the cast iron axle shaft for waterwheels).
- In 1782 he built the Chimney Mill in Britain.
- He also improved atmospheric engine, erecting one at Chasewater mine in Cornwall in 1775.

Key Events of Electrical Engineering

- **1820 – Ørsted's Discovery of Electromagnetism**
Hans Christian Ørsted discovered that an electric current creates a magnetic field.

- **1831 – Faraday's Law of Electromagnetic Induction**

- **1879 – Edison's Electric Light Bulb**

- **1888 – Tesla and AC Power**
Nikola Tesla demonstrated the feasibility of alternating current (AC) power transmission, which later became the dominant form of electrical power distribution.

Timeline of electronics

- **1904: Thermionic valve or diode invented.**
- **1907: Triode Patented**
- **1940: Ohl discovers that impurities in semiconductor crystals create photoelectric properties**
- **1947: First point contact transistor**
- **1952: First commercial device (hearing aid) to apply Shockley's junction transistor**
- **1954: First truly consistent mass-produced transistor is demonstrated**
- **1954: First transistor radio**

History of electronics

- **1955: Silicon oxide discovery**
- **1958-59: Integrated circuit invented**
- **1962: MOSFET is invented**
- **1966: Self aligned gate process for fabricating field effect transistor**
- **1967: First handheld calculator invented**
- **1969: Dynamic random access memory**
- **1970: The first CD ROM patented**
- **1971: Intel introduced computer on chip**

Industrial Engineering

- Industrial engineering (IE) is a branch of engineering that focuses on designing, improving, and managing systems that integrate people, materials, information, equipment, and energy. The goal of IE is to optimize these systems to improve productivity, efficiency, quality, and sustainability.

Key factors of Industrial Engineering

- **Man (Human Resources)** – Workforce involved in production and management.
- **Machine** – Equipment, tools, and technology used in manufacturing.
- **Material** – Raw materials, components, and supplies essential for production.
- **Method** – Standardized processes and best practices for efficiency and quality.
- **Measurement** – Performance metrics, quality control, and data analysis.
- **Money** – Financial resources, budgeting, and cost management.

- **Management** – Planning, organizing, and decision-making to optimize productivity.
- **Market** – Demand, competition, and customer needs that influence production.
- **Maintenance** – Upkeep of machines, equipment, and facilities to ensure smooth operations and prevent breakdowns.

Evolution of Industrial Engineering

Period	Main features	Main Contributors
1750–1850	Division of labor, replacing muscle by machine, concept of interchangeability	Adam Smith, Eli Whitney, Samual Bentham, Marc Brunel, Henry Maudslay and James Watt, Charles Babbage
1851–1930	Method study, distinct identity of industrial/production engineering, use of electrical power, steel production, SQC, queuing theory	Taylor, Gilbreth, Towne, Ford, Gantt, Shewhart, Erlang, Honer, Diemer
1931–1960	Ergonomics, quality management, operations research	Maslow, Mayo, McGregor, Deming, Juran, Dantzig, Walker, Kelley, Hamilton, U.S. Navy
1961–2010	Automation, quality, supply chain management, lean and agile manufacturing, six sigma, web based production, cloud computing and manufacturing	Taguchi, Ohno, Morley, Jobs, Gates
2010 onwards	Industry 4.0, Internet of things, sustainable manufacturing, big data and analytics, digital manufacturing	German Federal Ministry of Education and Research, European Union, companies like Google

Operations Research

- Operations Research (OR) is a **scientific and mathematical approach** to decision-making and problem-solving in complex systems. It involves using **optimization, statistics, mathematical modeling, and analytical methods** to improve efficiency, resource allocation, and performance in various industries such as manufacturing, logistics, healthcare, and finance.

Key topics in OR

- **Optimization** – Finding the best possible solution under given constraints.
- **Mathematical Modeling** – Representing real-world problems using equations and variables.
- **Simulation** – Imitating real-world operations to analyze different scenarios.
- **Statistics & Probability** – Managing uncertainty and making data-driven decisions.
- **Queuing Theory** – Studying waiting lines to optimize service efficiency.
- **Network Analysis** – Optimizing paths, routes, and connections in logistics and supply chains.

Evolution of OR

- Operations Research (OR) was formally developed **during World War II (1939–1945)** as military forces sought **scientific methods** to enhance decision-making and resource allocation. The war presented complex problems in logistics, strategy, and combat efficiency, which led to the systematic application of mathematical and analytical techniques.
- **1939-1945 (World War II)** – The **British and American military** extensively use Operations Research to optimize radar deployment, anti-submarine warfare, and logistics.
- **1947** – George Dantzig develops the **Simplex Method** for solving linear programming problems, revolutionizing optimization.